

REWEAR-FUSED: Techno-Economic & Life-Cycle Impact Dossier for Cleantech Judges

Prepared for an audience of Cox Enterprises executives and cleantech investors who speak in \$/ton CO₂ avoided, Scope 4, additionality, and ISO-grade LCA. Current as of mid-2026. All figures carry sources, units, and uncertainty bands. Models are explicitly labeled early-TRL/in-silico.

TL;DR

- **REWEAR-FUSED's single most defensible claim is a leverage story, not a bulk-tonnage story:** by enzymatically cleaving the ~2–5% elastane that today causes recyclers to reject an entire blended stretch garment, the platform unlocks fiber-to-fiber recycling of the remaining 95–98% (cotton/polyester). The defensible headline number is **~2 tonnes CO₂e avoided per tonne of blended stretch garment diverted** (range ~1.5–4 t, dominated by virgin-fiber displacement plus avoided landfill/incineration), and **~0.3 kg CO₂e per child's garment** (range 0.2–0.4).
- **The economics are well-precedented and enzyme cost is *not* the binding constraint.** NREL/BOTTLE's enzymatic-PET TEA shows recycled monomer reaching cost parity with virgin (recycled PET modeled at **\$1.51/kg vs \$1.87/kg virgin** in 2025), with **feedstock cost, solids loading and conversion yield — not enzyme cost — dominating**. REWEAR-FUSED's plausible abatement cost is **\$50–200/t CO₂ (expected case)**, best case negative (profitable recycling), worst case >\$500/t — competitive against direct air capture at ~\$600–1,000/t.
- **Near-term revenue is regulatory and exists before any plant is built.** California SB 707 (PRO = Landbell USA; producers must join by **July 1, 2026**) and EU textile EPR use **eco-modulated fees that penalize exactly the poly-cotton-elastane blends REWEAR-FUSED fixes**. The Digital Recyclability Passport + a recyclable-by-design elastane is a fee-reduction / malus-avoidance compliance product with immediate willingness-to-pay.

Key Findings

1. **Virgin elastane is among the most carbon-intensive mass-market fibers: ~15–20 kg CO₂e/kg** (a widely cited LCA benchmark gives ~17 kg CO₂e/kg at 70 dtex with cumulative energy demand ~380 MJ/kg; other sources cite ~20). It is a segmented polyurethane-urea built from carbon-intensive precursors — MDI ~4.3, TDI ~5.9, polyether polyol ~2.9–3.6 kg CO₂e/kg (ecoinvent 3.9/3.10). For comparison: virgin PET ~2.2 kg CO₂e/kg, cotton ~5.3, nylon 6 ~5–9.
2. **The "elastane problem" is quantified and severe.** An estimated 80% of clothing sold in the U.S. contained spandex in 2010, yet elastane is only ~1% of global fiber mass (1.2 Mt produced in 2022, ~2.8% recycled; ~1.5 Mt in 2024). Per Textiles Intelligence, "just 1% of elastane fibers in an otherwise all-cotton T-shirt is sufficient for the garment to be rejected at a recycling sorting plant"; TU Wien research shows ≥5% elastane causes clogging/jamming

of shredders, making mechanical recycling impossible. A trivial mass fraction strands the recyclable 95–99%.

3. **End-of-life today is value-destroying.** ~92 Mt/yr global textile waste (UNEP). Per Science Advances (2024), "Less than 1% of textile waste is recycled, with approximately 73% dumped in landfills or incinerated, 14% lost during production and collection, and 12% downcycled." Carter's KIDCYCLE (with TerraCycle) is explicit downcycling: synthetics → composite lumber/sheeting; mixed-fiber → mattress filler/pet bedding; cotton → insulation — terminal, not circular.
4. **The enzymatic-recycling LCA/TEA literature transfers directly.** NREL/BOTTLE 2021 Joule: rTPA reduces energy 69–83% and GHG 17–43% vs virgin TPA; 2025 Nature Chemical Engineering: recycled PET \$1.51/kg vs \$1.87/kg virgin, acid/base –99%, OpEx –74%. Carbios estimates ~–30% CO₂ vs an incineration/landfill EoL mix.
5. **REWEAR-FUSED's enzyme science is real and recent.** Carbamate-hydrolase / urethanase enzymes for polyurethane depolymerization are published (Branson et al. 2023, *Angew. Chem.*; ACS Catalysis 2025; chemo-enzymatic PU cascade depolymerization 2025–2026) — validating the (A) cleavable-elastane + (B) matched-enzyme co-design thesis, but at low TRL (~3–5).
6. **Policy creates willingness-to-pay now.** SB 707 (signed 2024; Landbell USA selected PRO Feb 27, 2026; join by July 1, 2026; eco-modulated fees with malus penalties for blended/hard-to-recycle products; full implementation ~2030; up to \$50,000/day penalties) and EU Waste Framework Directive textile EPR (mandatory collection from Jan 1, 2025; national EPR ~mid-2028).
7. **Cox's own thesis fits.** Cox led Nexus Circular's \$150M round (Jan 2023), became majority owner, and has surpassed \$3B in cleantech investment. Nexus uses advanced (pyrolysis) recycling to turn hard-to-recycle plastic into virgin-equivalent feedstock. REWEAR-FUSED is the textile-blend analogue: "Nexus for stretch textiles."

Details

PART 1 — Carbon & environmental baseline (what REWEAR-FUSED displaces)

(a) Virgin elastane / polyurethane footprint. Elastane is a segmented polyurethane-urea (≥85% polyurethane). Independent LCA benchmarks place cradle-to-gate global warming potential at ~15–20 kg CO₂e/kg — a benchmarking study widely cited in the trade reports ~17 kg CO₂e/kg (70 dtex) with cumulative energy demand ~380 MJ/kg (higher than polyester ~355 MJ/kg or acrylic ~360, comparable to nylon ~410 and cotton ~460); Géopélie cites ~20 kg CO₂e/kg. Precursor footprints from ecoinvent 3.9/3.10: MDI ~4.3, TDI ~5.9, isophorone diisocyanate ~6.3, HDI ~5.5, polyether polyol ~2.9–3.6 kg CO₂e/kg. Recycled-content elastane (Lycra EcoMade, Hyosung Creora Regen) reportedly cuts ~20–40% of CO₂ (~2.1 kg CO₂e/kg reduction) and ~45–55% of water, but capacity is <10,000 t/yr vs >100,000 t/yr virgin.

(b) End-of-life fates. A 2025 Australian textile-waste LCA found **100% landfilling of 1,000 kg textile waste generates ~8,310 kg CO₂e — the worst pathway**. A Swedish textile-recycling carbon-footprint study (Zamani et al.) reports recycling *credits* (avoided emissions): mechanical recycling of polyester/polyamide –11 to –4,295 kg CO₂e/tonne; chemical recycling of polyester ~–900 kg CO₂e/tonne; dissolution of polycotton ~–5,500 kg CO₂e/tonne; vs landfilling +12 to –139 and incineration with energy recovery ~0.23 kg CO₂e/tonne. **Carter's KIDCYCLE → TerraCycle is open-loop downcycling** (the company's own ESG site confirms shredding into mattress insulation, pet bedding, sports equipment) — it displaces only low-value virgin materials and is terminal.

(c) Textile-waste burden. ~92 Mt/yr global textile waste (UNEP; estimates 92–100 Mt, projected ~134 Mt by 2030); the sector is 8–10% of global GHG (more than aviation + shipping combined). Synthetic textiles are ~35% of primary microplastics entering oceans (IUCN). Microfiber shedding (PLOS One 2021): 9.6–1,240 mg/kg textile per wash (~8,809 to >6.8M fibers per wash); a single load can shed 700,000–12M fibers; elastane-blended fabrics complicate shedding and recycling.

(d) The elastane problem, quantified. Elastane in >80% of clothing but ~1% of fiber mass (1.2 Mt in 2022, ~2.8% recycled). The "killer fiber" mechanism: 1% elastane → whole-garment rejection at sorting; ≥5% → mechanically unrecyclable. This is REWEAR-FUSED's exact leverage point — the elastane fraction is a contamination "master key" that strands the recyclable cotton/polyester majority.

(e) Virgin-material displacement value. Recovered fibers/monomers displace virgin production. Virgin spandex sells multi-thousand \$/tonne; recycled elastane commands a ~10–20% premium. Recovered cotton/polyester displaces virgin polyester (~\$1.0–1.9/kg) and cotton. Market context: **textile-to-textile recycling ~\$2.8B (2024) → ~\$44.8B (2034) at 24.9% CAGR** (Global Market Insights); biological/enzymatic recycling is a named sub-segment; broader textile recycling ~\$8.41B (2025) → ~\$11.88B (2030) at 7.2% (MarketsandMarkets). [GlobeNewswire](#)

PART 2 — LCA of enzymatic textile recycling (the REWEAR-FUSED pathway)

(a)/(c) Published enzymatic-recycling LCAs. The anchor body of work is NREL/BOTTLE on enzymatic PET. The **2021 Joule TEA/LCA** (Singh et al., DOI 10.1016/j.joule.2021.06.015): enzymatic rTPA reduces supply-chain energy **69–83%** and GHG **17–43%** per kg vs virgin TPA; economy-wide up to 95% impact reduction. The rigorous process-based **Green Chemistry 2022 LCA** (Uekert et al., DOI 10.1039/D2GC02162E) provides the essential honesty check: **as-modeled (TRL ~5), enzymatic hydrolysis performs 1.2–17× worse than virgin across most impact categories** (top contributors: post-consumer collection/processing, sodium hydroxide, electricity; ~3× higher water use), reaching parity with virgin only after improving yields and eliminating amorphization pretreatment and pH control. **Methodological takeaway for judges: enzymatic recycling is not automatically greener — the carbon case rests on displaced virgin production + avoided landfill/incineration, plus designed-in yield/energy improvements, not on the process being intrinsically low-impact.**

(b) Process burden. Drivers: enzyme-production footprint, depolymerization energy (mild ~60 °C for PET hydrolases; PU/elastane urethanases likely need thermal pretreatment ~180 °C per 2025–2026 cascade-depolymerization work), monomer separation/purification (crystallization, distillation), and re-polymerization. The 2025 Nature Chem Eng process cut acid/base 99%, OpEx 74%, and ethylene-glycol-recovery distillation energy 65% vs the prior enzymatic process — demonstrating the process-impact gap is closeable. [Wiley Online Library](#)

(d) Methodologies/standards to cite. ISO 14040:2006 (principles & framework) and ISO 14044:2006 (requirements & guidelines); EU Product Environmental Footprint and the **PEFCR for Apparel & Footwear (approved by the European Commission in 2025; Technical Secretariat coordinated by Cascale)**; Higg Materials Sustainability Index (Higg MSI) and Higg Product Module (Cascale/Worldly); GHG Protocol Product and Project standards. Functional unit: "one day of wear" (PEFCR) for garment-level, or 1 kg fiber for material comparisons. [ResearchGate](#)

(e) Data/tools. ecoinvent (v3.9/3.10), Higg MSI, GaBi/Sphera, SimaPro, OpenLCA, Textile Exchange data. NREL's PET LCA template — SimaPro + ecoinvent 3.3 + TRACI 2.1 + AWARE (water) — is directly citable.

PART 3 — Techno-economic analysis (cost & unit economics)

(a)/(b) Cost structure & published TEA (precise figures). NREL Joule 2021 base case: rTPA **minimum selling price \$1.93/kg**, "dominated by feedstock cost (assuming a PET flake price of \$0.66/kg)"; modeled facility **150 metric tons/day, total capital investment ~\$67M, OpEx ~\$44M/yr**; enzyme base-case cost **\$15/kg** (literature range \$25–110/kg). **Crucially, "Sensitivity analyses indicated that variability of the MSP of rTPA was most affected by solids loading and extent of conversion. Enzyme loading, enzyme cost, and reaction residence time were less impactful"** (enzyme loading/residence time move MSP only ~5–9%). The **2025 Nature Chem Eng** process: recycled PET **\$1.51/kg vs \$1.87/kg virgin** (cost parity/advantage) via >99% acid/base reduction and 74% OpEx reduction. *(Note: these are PET figures; elastane's carbamate chemistry needs different enzymes and likely thermal pretreatment, which would raise process cost/energy vs the PET base case.)*

(c) Displaced alternatives' economics. Virgin spandex multi-thousand \$/tonne; avoided landfill tipping + incineration costs; **avoided SB 707 eco-modulated malus fees** (which specifically target mixed-material blends like poly-cotton-elastane); EU EPR (France Refashion model) eco-modulation bonuses/penalties.

(d) Digital Recyclability Passport as near-term revenue. SB 707 and the forthcoming EU Digital Product Passport require SKU-level fiber-composition and recyclability data; compliance consultants explicitly note DPPs are "rapidly becoming the most efficient way to prove fiber composition and secure lower 'Bonus' Eco-Modulated fees." REWEAR-FUSED's passport + recyclable-by-design elastane is a compliance product with willingness-to-pay *before* recycling plants exist — the asset-light wedge.

(e) \$/ton CO₂ abatement. Because enzymatic monomer recovery can approach or beat virgin cost, REWEAR-FUSED's pathway abatement cost is plausibly **\$50–200/t CO₂ (expected)**, best case negative (profitable recycling generating saleable fiber), worst case >\$500/t (high enzyme cost + low recovery). Benchmarks: direct air capture ~\$600–1,000/t by 2030 (Herzog/MIT via Yale E360; IEA scaled estimates \$230–630/t; aspirational \$100–200 long-term); U.S. sectoral marginal abatement costs span \$27–\$16,899/t, with most sectors <\$500/t and a \$50–500/t carbon price unlocking ~52% of emissions. **An expected <\$200/t pathway with a co-product revenue stream is a strong cleantech number.**

PART 4 — Cleantech-investor metrics & framing

(a) Scope 4 / avoided emissions. Use the **WBCSD Guidance on Avoided Emissions v2.0 (July 2025)** — the reference methodology — with its three eligibility gates (climate-action credibility, climate-science alignment, contribution legitimacy). The concept originates from WRI's 2013 working paper "Estimating and Reporting the Comparative Emissions Impacts of Products." Compute as: (baseline: garment landfilled/incinerated + virgin fiber produced) – (REWEAR-FUSED: enzymatic recycling + recovered fiber). **Per SBTi, avoided emissions must be reported separately from Scope 1/2/3** — never net them against your own inventory. GHG Protocol Project Accounting and the 2024 "Beyond Value Chain Mitigation" consultation support project-level claims. The Avoided Emissions Platform (Quantis/Robeco/WBCSD) offers a standardized factor approach.

(b) Additionality (three legs). (i) **Material additionality** — blended stretch garments are categorically non-recyclable today ($\geq 1\%$ elastane $\rightarrow$ rejection), so this waste is landfilled/incinerated/downcycled by default; recycling it is incremental, not business-as-usual. (ii) **Technological additionality** — no commercial enzyme cleaves elastane carbamate bonds at scale (urethanases are TRL ~3–5), so the recovered fiber would not otherwise be recovered. (iii) **Policy additionality** — REWEAR-FUSED enables recyclable-by-design outcomes that fast-fashion design economics would not produce unaided.

(c) Market size / impact at scale. TAM: textile-to-textile recycling ~\$44.8B by 2034. SAM: blended children's stretch apparel within the ~1.2–1.5 Mt/yr elastane-containing stream. At scale, diverting blended stretch garments to fiber-to-fiber recycling abates ~2 t CO₂e per tonne; a 100,000 t/yr program $\approx$ **~200,000 t CO₂e/yr avoided** plus ~100,000 t waste diverted and proportional microfiber/landfill reductions.

(d) Benchmarks. "Good" cleantech abatement is broadly **<\$100–200/t CO₂** with a credible scale path and a co-product revenue stream that de-risks the carbon thesis. Capital intensity is the watch-item (NREL's 150 t/day plant $\approx$ \$67M CapEx). Frame an **asset-light early phase** — Digital Recyclability Passport SaaS + licensing the cleavable-fiber/enzyme IP to fiber producers — ahead of capital-heavy depolymerization plants.

(e) Cox/Nexus framing. Speak Cox's language directly: Nexus Circular converts hard-to-recycle plastic into ISCC-PLUS-certified virgin-equivalent feedstock via advanced recycling; Cox

invested \$150M (2023), is majority owner, and has surpassed \$3B in cleantech. **REWEAR-FUSED** = "Nexus for stretch textiles" — biological recycling of the blended feedstock pyrolysis can't cleanly address — and mirror Nexus's commercial-demonstration-to-scale path.

PART 5 — Integrated impact model & the slide numbers

Bottom-up model (defensible, conservative):

- **Per garment** (child's stretch item ~0.15 kg, ~5% elastane): diverting from landfill/downcycling to fiber-to-fiber recycling avoids ~**0.2-0.4 kg CO₂e** (expected ~0.3), from virgin-fiber displacement + avoided EoL.
- **Per tonne of blended stretch garment: ~1.5-4 t CO₂e avoided** (expected ~2 t), dominated by displaced virgin fiber — consistent with literature recycling credits (polycotton mechanical -0.45 to -3.3 t/t; polyester chemical ~-0.9 t/t; polycotton dissolution ~-5.5 t/t).
- **At scale (100,000 t/yr diversion): ~200,000 t CO₂e/yr avoided**, ~100,000 t waste diverted.
- **Abatement cost: \$50-200/t CO₂ expected**; best case negative; worst case >\$500/t.

Sensitivity drivers (rank order, from the NREL TEA analogy): (1) feedstock collection/sorting cost & garment recovery rate; (2) depolymerization solids loading & conversion yield; (3) monomer recovery/purity; (4) plant scale & grid carbon intensity; (5) enzyme cost/loading (minor — explicitly de-risked by NREL's finding). The **carbon result is most sensitive to (a) the EoL baseline displaced** (NREL found switching an 80/20 landfill/incineration mix to 100% incineration changes impacts up to +30%) and **(b) the virgin-fiber displacement credit**.

Best / expected / worst cases:

Metric	Worst	Expected	Best
CO ₂ e avoided per tonne garment	~1.5 t	~2 t	~4 t
Abatement cost	>\$500/t	\$50-200/t	negative (profitable)
Recovered-fiber value	displaces low-value downcycle	displaces virgin polyester/cotton	displaces virgin elastane (premium)

Honest TRL framing. Present numbers as an **in-silico, early-TRL (~3-4) impact model**, not measured performance. State explicitly: (i) the as-modeled enzymatic process can be worse than virgin until yields/pretreatment are optimized (Uekert 2022); (ii) urethanase enzymes are pre-commercial; (iii) the credible near-term value is the passport/compliance layer + design-for-recyclability, with the full closed-loop carbon benefit realized only at scale. Always show best/worst/expected bands and cite WBCSD v2.0 + ISO 14040/44 + EU PEF CR for method credibility.

Recommendations

1. **Lead with the leverage ratio, not bulk tonnage.** "2–5% elastane strands 95–98% of the garment; we cleave the 2–5% and unlock the rest." Pair with one headline: **~2 t CO₂e avoided per tonne diverted (range 1.5–4)** and **<\$200/t CO₂ expected abatement** vs DAC ~\$600–1,000/t. *Threshold to upgrade the claim from modeled to demonstrated:* pilot-scale conversion-yield and fiber-recovery-rate data on a real cotton/polyester/elastane garment.
2. **Sell the compliance layer first.** Position the Digital Recyclability Passport as the SB 707 / EU-DPP fee-reduction product generating revenue before plants exist. Anchor on hard dates: producers must join Landbell USA by **July 1, 2026**; eco-modulated malus fees target exactly the poly-cotton-elastane blends REWEAR-FUSED fixes. *Threshold that raises willingness-to-pay:* publication of the SB 707 stewardship-plan fee schedule (expected ~2029) and EU eco-modulation criteria.
3. **Adopt a citable method stack today.** ISO 14040/14044 + EU PEFCR for Apparel & Footwear (EC-approved 2025) + Higg MSI + WBCSD Avoided Emissions v2.0; model in SimaPro/OpenLCA on ecoinvent 3.10 with TRACI/AWARE; report avoided emissions separately from Scope 1/2/3 per SBTi. This pre-empts the "hand-waving" critique outright.
4. **Frame for Cox specifically** as "Nexus for stretch textiles" — biological/advanced recycling of hard-to-recycle feedstock into virgin-equivalent fiber — and mirror Nexus's asset-light-to-capital-heavy scaling path and ISCC-PLUS-style certification.
5. **Pre-register the sensitivity analysis** with best/worst/expected bands. Never present a single point estimate to LCA-literate judges; show the bands and name the top three drivers (recovery rate, conversion yield, displaced baseline).

Caveats

- The ~15–20 kg CO₂e/kg elastane figure comes from secondary LCA benchmarks (and some non-primary popular sources like Géopélie ~20), not a single peer-reviewed primary dataset — flag and triangulate before publishing a point value.
- NREL/BOTTLE figures are for **PET**, not elastane/polyurethane. They transfer directionally, but elastane's carbamate chemistry requires different (urethanase) enzymes and likely thermal pretreatment, which would raise process energy/cost vs the PET base case. State this explicitly.
- The process-based LCA (Uekert 2022) shows enzymatic recycling can be worse than virgin until optimized; the carbon case depends on displacement credits and yield improvements, not the process being intrinsically clean.
- The abatement-cost range (\$50–200/t expected) is modeled by analogy to NREL PET economics, not a REWEAR-FUSED-specific TEA; label it illustrative early-stage.
- **Market-timing context:** the LYCRA Company filed Chapter 11 on March 17, 2026 — signaling incumbent elastane-industry fragility (an opportunity narrative, but also

feedstock-supply uncertainty).

- SB 707 fee schedules are not yet finalized (eco-modulated fees likely invoiced ~2029–2030; AAFA filed a legal challenge to the Landbell selection on March 27, 2026), so EPR-fee-avoidance willingness-to-pay is directional, not yet a fixed \$/garment.
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Source quick-reference (working URLs)

- Elastane footprint: impactful.ninja/how-sustainable-are-elastane-fabrics; yulex.com/post/rubber-chronicle-25; PU precursors via RSC D5GC00423C (pubs.rsc.org/en/content/articlehtml/2025/gc/d5gc00423c).
- Elastane recycling problem: textile-network.com/en/Business/Barriers-in-the-recycling-loop-threaten-existence; pmc.ncbi.nlm.nih.gov/articles/PMC11767914; en.wikipedia.org/wiki/Spandex.
- Textile-waste / EoL: UNEP via sustainableatlas.org; Science Advances "Chemical recycling of mixed textile waste" (doi 10.1126/sciadv.ado6827); Australian LCA pmc.ncbi.nlm.nih.gov/articles/PMC11953081; Swedish credits via ResearchGate 267812377.
- Carter's KIDCYCLE: ir.carters.com/news-releases (Apr 13, 2021); esg.carters.com/waste; terracycle.com/en-US/brigades/carters.
- NREL TEA/LCA: Joule 2021 ([cell.com/joule/fulltext/S2542-4351\(21\)00303-2](https://cell.com/joule/fulltext/S2542-4351(21)00303-2); DOI 10.1016/j.joule.2021.06.015); Green Chemistry 2022 (pubs.rsc.org/en/content/articlehtml/2022/gc/d2gc02162e; DOI 10.1039/D2GC02162E); Nature Chem Eng 2025 (nature.com/articles/s44286-025-00212-y; DOI 10.1038/s44286-025-00212-y).
- Urethanase enzymes: Branson et al. 2023 (onlinelibrary.wiley.com/doi/10.1002/anie.202216220); ACS Catalysis 2025 (pubs.acs.org/doi/10.1021/acscatal.5c01062); PU cascade depolymerization (chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/cssc.202502633).
- Methods: ISO 14040/14044; EU PEFCR Apparel & Footwear (cascale.org/about-us/advocacy/pef); Higg MSI methodology (howtohigg.cascale.org).
- Avoided emissions: WBCSD v2.0 (wbcsd.org/resources/guidance-on-avoided-emissions...); net0.com/blog/scope-4.
- SB 707: dlapiper.com (Feb/Oct 2025 + 2026); anthesisgroup.com/insights/sb-707; americancirculartextiles.com/sb-707-implementation-tracker.
- Cox/Nexus: coxenterprises.com/news/nexus-circular-secures-150-million-funding-round; nexuscircular.com.
- DAC benchmarks: belfercenter.org (Prospects for DAC); Yale E360 / Herzog.
- Market size: gminsights.com/industry-analysis/textile-to-textile-recycling-market; marketsandmarkets.com/textile-recycling.